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ICT Corner

Properties of Gases

By using this tool you will be able to understand the relationship between T, P of V gases and verify the gas laws.

Please go to the URL
https://phet.colorado.edu/en/simulation/gas-properties
(or)
Scan the QR code on the right side

B164_11_CHE_EM

Step – 1

Open the Browser and type the URL given (or) Scan the QR Code. In the website click the gas properties applet, then you will see a Java applet named 'gas properties' opening up as shown in the figure.

Set up instructions:

- In the chamber (2) you can pump some gas molecules by using the pump handle (5).
- The temperature (1) and pressure (6) of the gas molecules in the chamber can be monitored by using the corresponding gauges.
- The volume of the chamber can be altered by moving the left side wall (3).
- The temperature of the chamber can be altered by heating or cooling using the slider (7)
- Any of the three properties can be kept as constant by making appropriate selections in the box (4)

Verifying the gas laws:

By using the above instructions you can understand the gas laws. To understand the Boyle's law, keep the temperature constant (4) and pump few molecules of gas (5) and keep it as constant. Now reduce the volume by pushing the left side wall (3) then you can see that the pressure of the gas increases. Similarly the other two gas laws can also be verified.

